Sales Data Analysis & Visualization | Python Jupyter Notebook

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In [2]:
```
# Let's import python libraries First

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # visualizing data
%matplotlib inline
import seaborn as sns
```

In [7]:
```
# import csv file
df = pd.read_csv('Sales_Data.csv', encoding= 'unicode_escape')
```

In [8]:
```
#Check Number of rows and Columns
df.shape
```

Out[8]:
```
(11257, 13)
```

In [9]:
```
#To see the imported data
df.head(5)  # Show Top 5 Rows
```

Out[9]:

	User_ID	Cust_name	Product_ID	Age	Age Group	Gender	State	Zone	Zi
0	1002903.0	Ajay	P00125942	27.0	26-35	Female	Maharashtra	West	
1	1000732.0	Vikas	P00110942	34.0	26-35	Female	Andhra Pradesh	South	
2	1001990.0	Sunil	P00118542	16.0	0-17	Female	Uttar Pradesh	Central	
3	1001425.0	Virendra	P00237842	16.0	0-17	M	Karnataka	South	
4	1000588.0	Vishal	P00057942	28.0	26-35	M	Gujarat	West	

In [20]:
```
#Field details and Data type
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11257 entries, 0 to 11256
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   User_ID           11245 non-null  float64
 1   Cust_name         11245 non-null  object
 2   Product_ID        11245 non-null  object
 3   Age               11245 non-null  float64
 4   Age Group         11245 non-null  object
 5   Gender            11245 non-null  object
 6   State             11245 non-null  object
 7   Zone              11245 non-null  object
 8   Zipcode           0 non-null      float64
 9   Profession        11245 non-null  object
 10  Product_Category  11245 non-null  object
 11  Orders            11245 non-null  float64
 12  Amount            11245 non-null  float64
dtypes: float64(5), object(8)
memory usage: 1.1+ MB
```

Let's Start Data Cleaning

In [21]:
```
#Deleting blank column
df.drop(['Zipcode'], axis=1, inplace=True)
```

In [22]:
```
#List of Columns Available
df.columns
```

Out[22]:
```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Age', 'Age Group', 'Gender',
       'State', 'Zone', 'Profession', 'Product_Category', 'Orders', 'Amount'],
      dtype='object')
```

In [23]:
```
#check for null values
pd.isnull(df).sum()
```

Out[23]:
```
User_ID             12
Cust_name           12
Product_ID          12
Age                 12
Age Group           12
Gender              12
State               12
Zone                12
Profession          12
Product_Category    12
Orders              12
Amount              12
dtype: int64
```

In [24]:
```
# drop null values
df.dropna(how='all', inplace=True)
```

In [25]:
```
df.shape
```

Out[25]:
```
(11245, 12)
```

In [16]:
```
#replace value of Gender Column
df['Gender'] = df['Gender'].replace('M', 'Male')
```

In [26]:
```
#View only Male Gender data
df[df['Gender'] == 'Male']
```

Out[26]:

	User_ID	Cust_name	Product_ID	Age	Age Group	Gender	State	Zone
3	1001425.0	Virendra	P00237842	16.0	0-17	Male	Karnataka	South
4	1000588.0	Vishal	P00057942	28.0	26-35	Male	Gujarat	West
5	1000588.0	Soni	P00057942	28.0	26-35	Male	Himachal Pradesh	Northern
8	1003224.0	Kushal	P00205642	35.0	26-35	Male	Uttar Pradesh	Central
11	1003829.0	Harshita	P00200842	34.0	26-35	Male	Delhi	Central
...	...	...	...	...	...	...	...	...
11249	1005446.0	Sheetal	P00297742	53.0	51-55	Male	Gujarat	West
11250	1005446.0	Sheetal	P00297742	53.0	51-55	Male	Madhya Pradesh	Central
11252	1000695.0	Manning	P00296942	19.0	18-25	Male	Maharashtra	West
11253	1004089.0	Reichenbach	P00171342	33.0	26-35	Male	Haryana	Northern
11255	1004023.0	Noonan	P00059442	37.0	36-45	Male	Karnataka	South

3408 rows × 12 columns

--> Now Let's Learn EDA - Exploratory Data Analysis

In [18]:
```
# describe() method returns description of the data in the DataFrame (i.e. count
df.describe()
```

Out[18]:

	User_ID	Age	Zipcode	Orders	Amount
count	1.124500e+04	11245.000000	0.0	11245.000000	11245.000000
mean	1.003004e+06	35.415651	NaN	3.500311	9461.934237
std	1.716207e+03	12.756369	NaN	1.713706	5234.426634
min	1.000001e+06	12.000000	NaN	1.000000	188.000000
25%	1.001492e+06	27.000000	NaN	2.000000	5443.000000
50%	1.003065e+06	33.000000	NaN	4.000000	8109.000000
75%	1.004429e+06	43.000000	NaN	5.000000	12683.000000
max	1.006040e+06	92.000000	NaN	6.000000	29350.000000

In [19]:

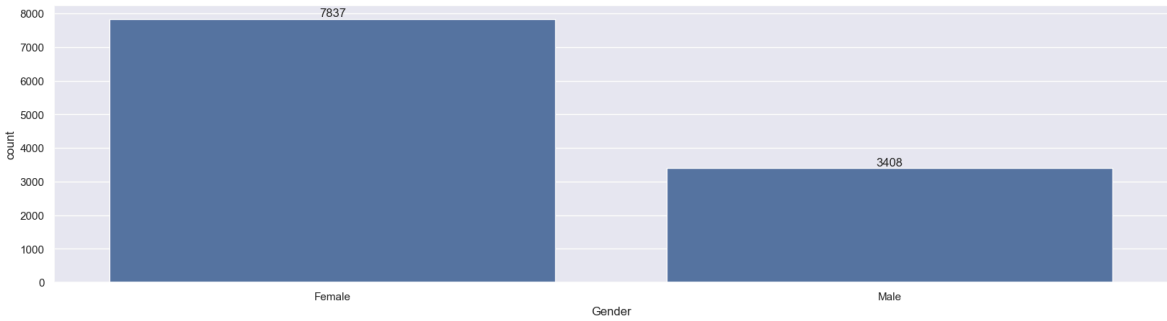
```
# use describe() for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

Out[19]:

	Age	Orders	Amount
count	11245.000000	11245.000000	11245.000000
mean	35.415651	3.500311	9461.934237
std	12.756369	1.713706	5234.426634
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	4.000000	8109.000000
75%	43.000000	5.000000	12683.000000
max	92.000000	6.000000	29350.000000

In [86]:

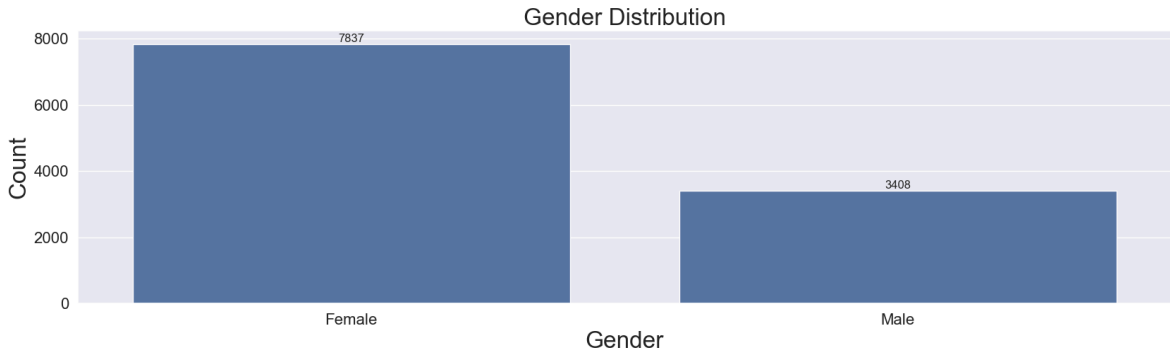
```
ax=sns.countplot(x='Gender', data = df)
for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```



In [89]:

```
# Total Transactions count by Gender Wise in Bar Chart
ax = sns.countplot(x = 'Gender', data = df)

# Set Title and Labels with Font Size
for bars in ax.containers:
    ax.bar_label(bars)
    ax.set_title('Gender Distribution', fontsize=24)   # Chart Title
```

```
    ax.set_xlabel('Gender', fontsize=24)              # X - Axis Label
    ax.set_ylabel('Count', fontsize=24)               # Y - Axis Label
    ax.tick_params(axis='both', labelsize=16)         # Axis Parameters
plt.show()
```

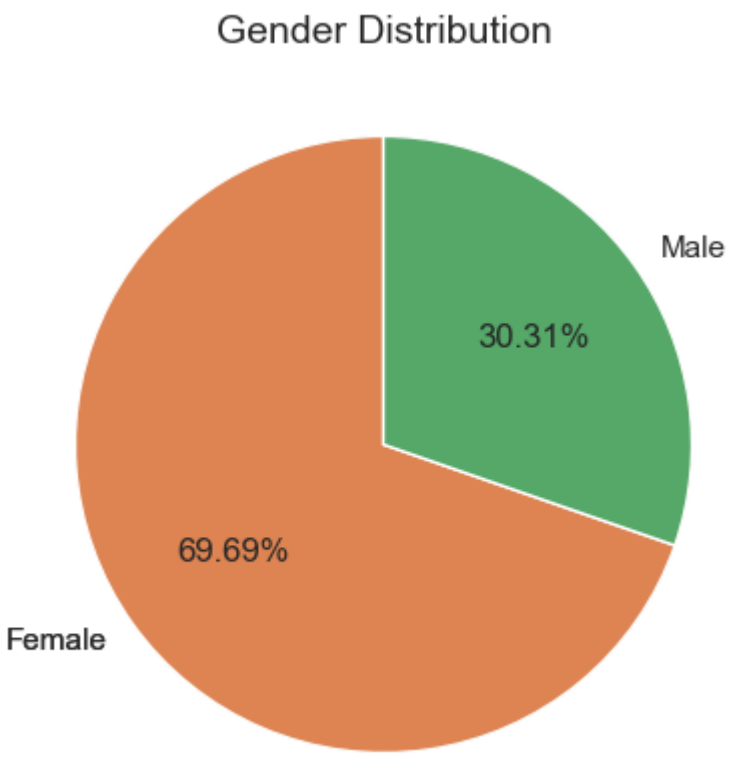


In [ ]:
```
# 2 . Gender wise distribution in Pie Chart
```

In [90]:
```
#Total Transactions Count by Gender wise
gender_counts = df['Gender'].value_counts()
print(gender_counts)
```

```
Gender
Female    7837
Male      3408
Name: count, dtype: int64
```

In [98]:
```
#Total Transactions Count by Gender wise in Pie Chart
gender_counts = df['Gender'].value_counts()

plt.pie(
    gender_counts,                        # Data (counts)
    labels=gender_counts.index,           # Labels (e.g., Male, Female)
    autopct='%.2f%%',                     # Show percentages (2 decimal place)
    startangle=90,                        # Rotate for better orientation
)
# Add a title
plt.title('Gender Distribution', fontsize=14)
plt.show()
```



In [104…

```
df.groupby('Gender', as_index=False)['Amount'].sum()
```

Out[104…

	Gender	Amount
0	Female	74461610.49
1	Male	31937840.00

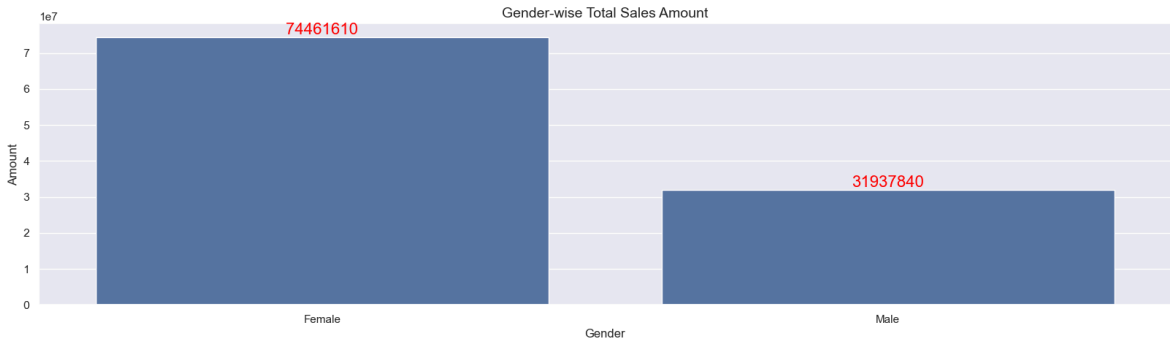
In [108…

```
Gen_Wise_Sales = df.groupby('Gender', as_index=False)['Amount'].sum().sort_value
print(Gen_Wise_Sales)
```

```
  Gender       Amount
0  Female  74461610.49
1    Male  31937840.00
```

In [113…

```
# Gender wise Total Sales Amount

# Step 1: Group data by Gender and sum the Amount
Gen_Wise_Sales = df.groupby('Gender', as_index=False)['Amount'].sum().sort_value

# Step 2: Create the bar plot
ax = sns.barplot(x='Gender', y='Amount', data=Gen_Wise_Sales)

# Step 3: Add data labels on top of bars
for bar in ax.containers:
    ax.bar_label(bar, fmt='%.0f', fontsize=16, color='red')  # Show counts as wh

# Step 4: Add titles
plt.title('Gender-wise Total Sales Amount', fontsize=14)
plt.show()
```

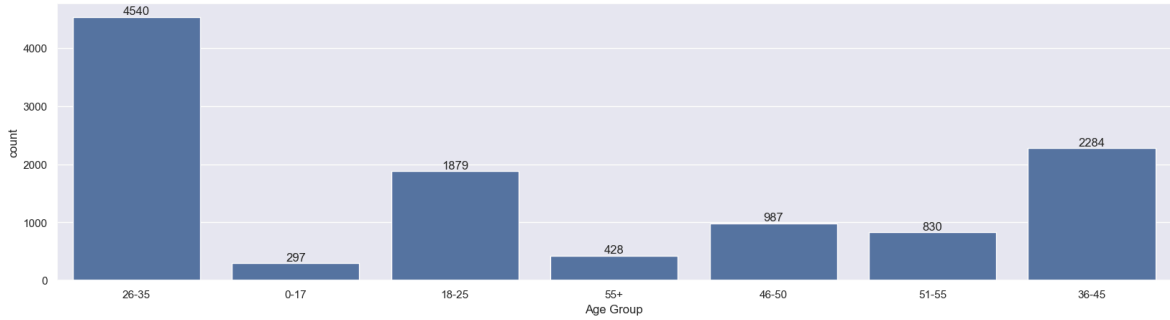


Conclusion : Male sales are higher than female sales

--> Age group wise total transactions

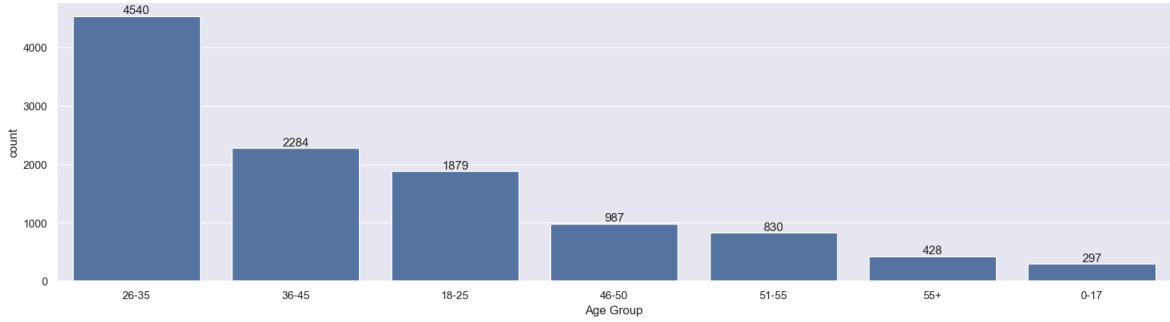
In [115…

```
#Age Group wise Transactions Count in Bar Chart
ax = sns.countplot(data = df, x = 'Age Group')

for bars in ax.containers:          #Show Data labels
    ax.bar_label(bars)
plt.show()
```



In [121…

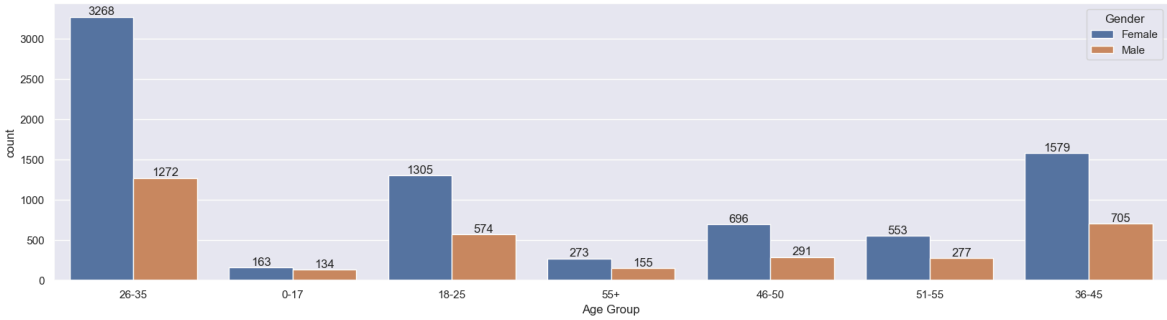
```
#Step 1 - Age group wise transactions count (Sorting)

age_group_counts=df['Age Group'].value_counts().sort_values(ascending=False)

#Step 2 - Use Ordered categories for sorting in countplot
sns_order=age_group_counts.index

#Age Group wise Transactions Count in Bar Chart
ax = sns.countplot(data = df, x = 'Age Group', order=sns_order)

for bars in ax.containers:          #Show Data labels
    ax.bar_label(bars)
plt.show()
```



In [123…

```
#Age group and Gender wise Transactions distribution
```

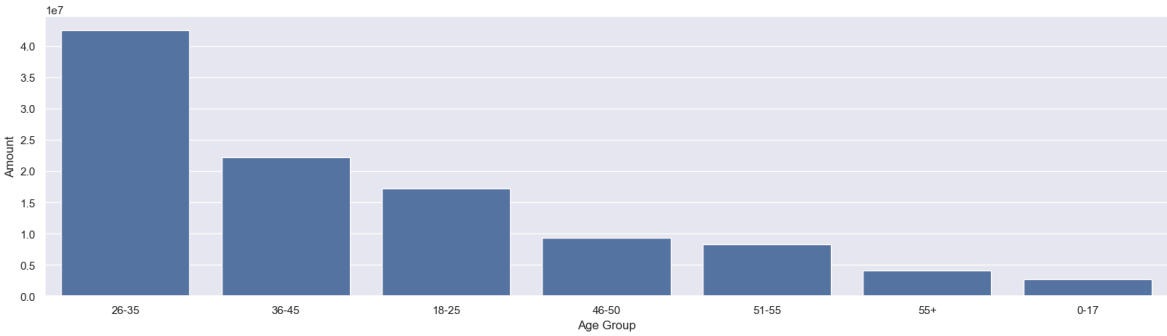
```
ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')

for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```



From above graphs we can see that most of the buyers are of age group between 26-35 yrs female

In [127…

```
#  Age Group wise Total Amount
sales_age = df.groupby(['Age Group'], as_index=False)['Amount'].sum().sort_value

sns.barplot(x = 'Age Group',y= 'Amount' ,data = sales_age)

plt.show()
```



--> State wise analysis

In [131…

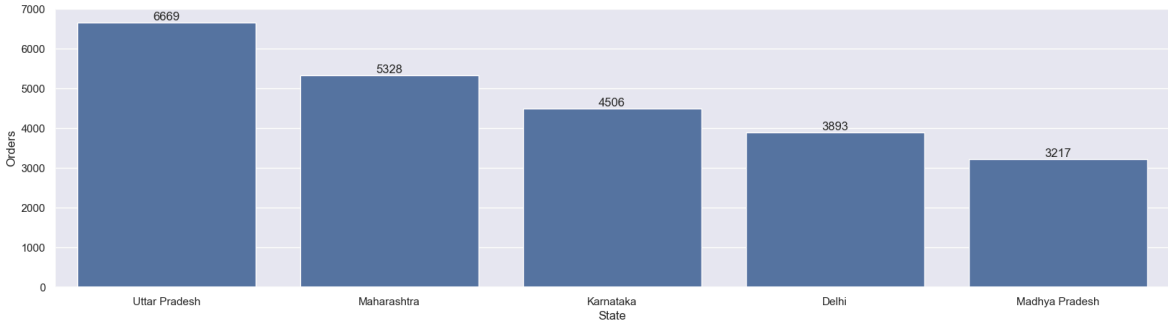
```
order_state = df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(
print(order_state)
```

```
            State  Orders
14   Uttar Pradesh  6669.0
10     Maharashtra  5328.0
7        Karnataka  4506.0
2            Delhi  3893.0
9   Madhya Pradesh  3217.0
```

In [133…

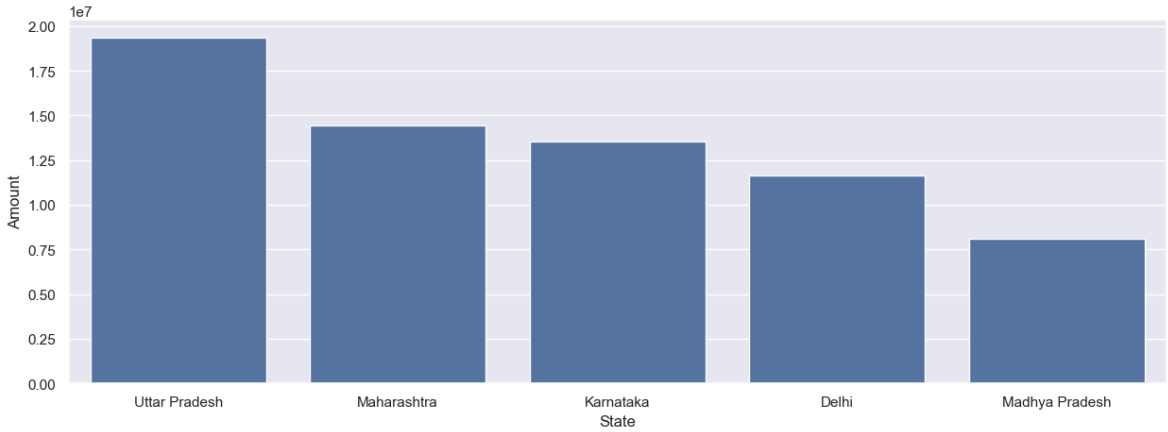
```
# Order wise Top 5 state
order_state = df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(

#sns.set(rc={'figure.figsize':(15,5)})
ax = sns.barplot(data = order_state, x = 'State',y= 'Orders')

for bars in ax.containers:
    ax.bar_label(bars)
```

```
plt.show()
```



In [138…

```
# Amount Wise top 5 state

sales_state = df.groupby(['State'], as_index=False)['Amount'].sum().sort_values(

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Amount')
plt.show()
```

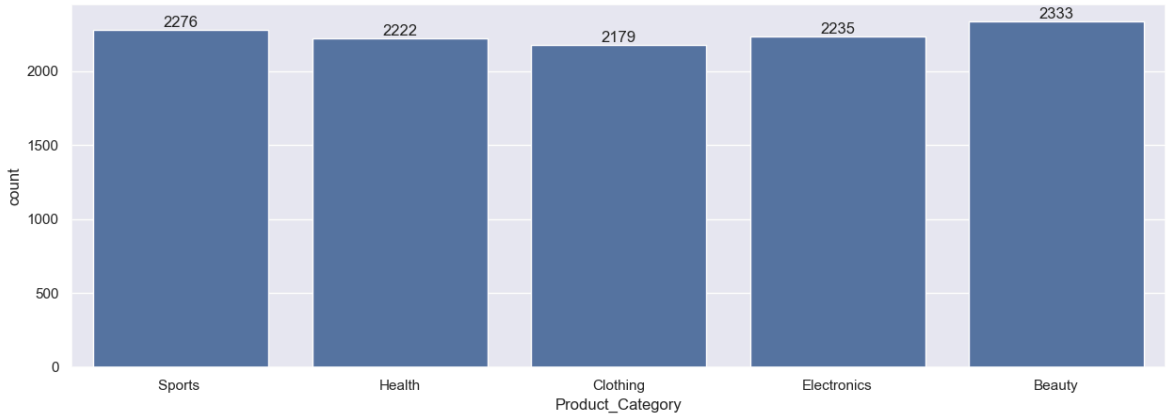


From above graphs we can see that most of the orders & total sales/amount are from Uttar Pradesh, Maharashtra and Karnataka respectively

--> Product_Category Analysis

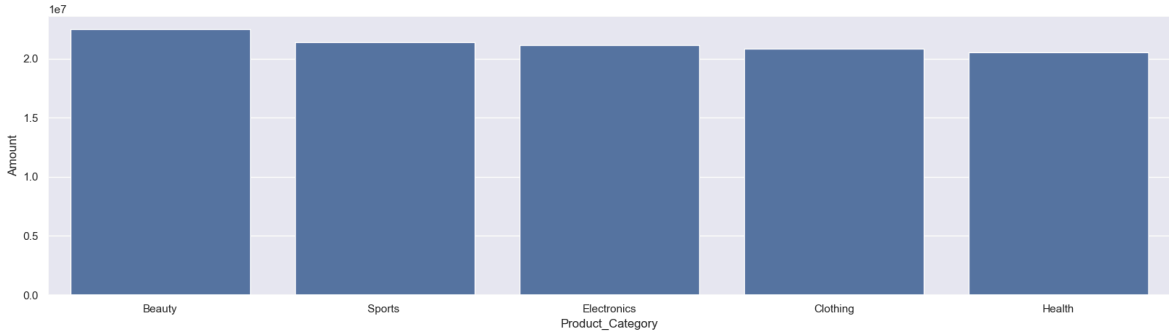
In [139…

```
# Product category wise Transactions count
ax = sns.countplot(data = df, x = 'Product_Category')

sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
plt.show()
```



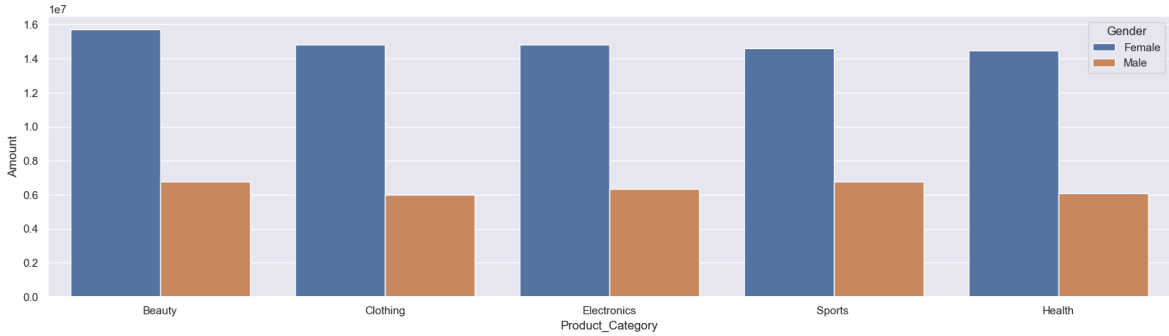
In [140…

```
#Amount wise Product Category in Bar Chart
sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().s

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount')

plt.show()
```



In [141…

```
#Product category and Gender wise Transactions count
sales_pro = df.groupby(['Product_Category', 'Gender'], as_index=False)['Amount']
sns.barplot(data = sales_pro, x = 'Product_Category',y= 'Amount', hue='Gender')

plt.show()
```



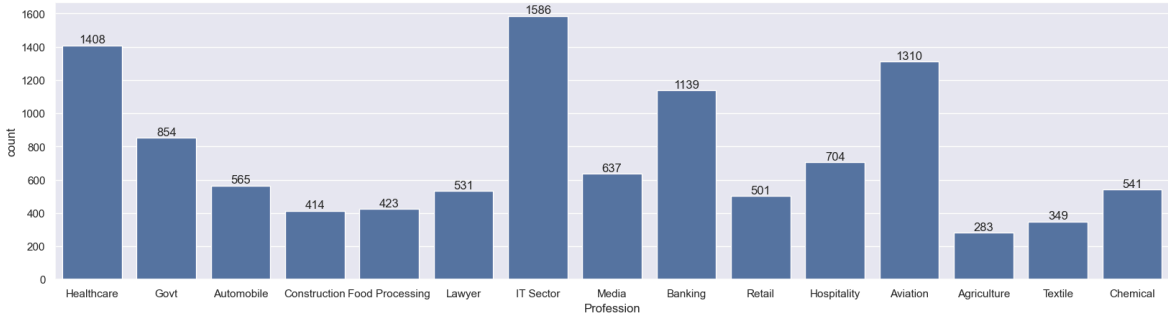
*From above graphs we can see that most of the buyers are Female

--> Profession wise Analysis

In [142…

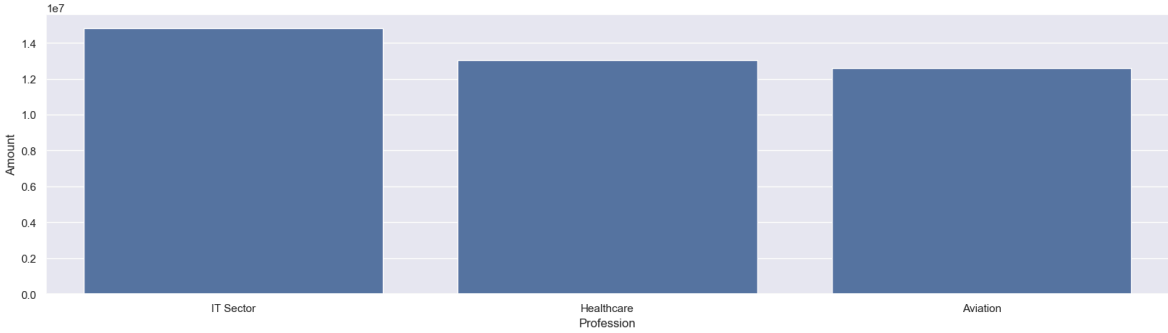
```
# Profession wise Transactions count

ax = sns.countplot(data = df, x = 'Profession')
for bars in ax.containers:
```

```
    ax.bar_label(bars)
plt.show()
```



In [147…

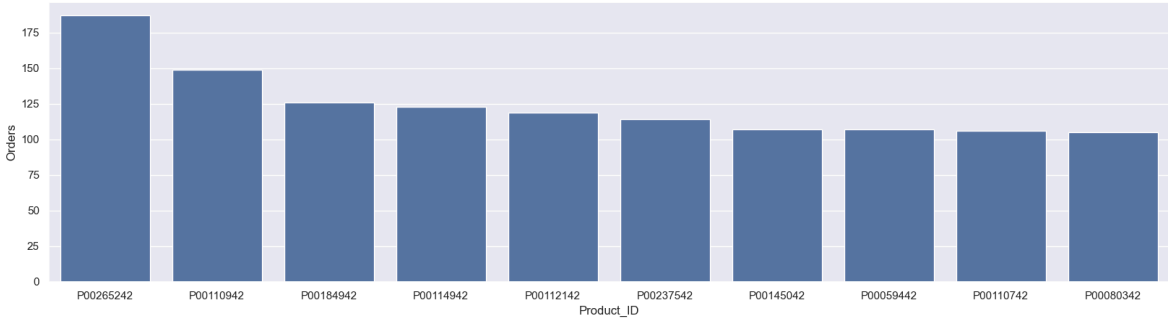
```
# Amount Wise Top Professions
sales_state = df.groupby(['Profession'], as_index=False)['Amount'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Profession',y= 'Amount')
plt.show()
```



From above graphs we can see that most of the buyers are working in IT, Healthcare and Aviation sector

In [148…

```
#Order wise Top 10 Product
sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_va

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
plt.show()
```



Conclusion:

Female age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Beauty, Sports and Electronics